



```
In [1]: print("Hello, World!")
```

Hello, World!

```
In [2]: a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("Sum =", a + b)
```

Sum = 14

```
In [5]: name = input("enter your name")
print("my name is",name)
```

my name is ishita dutta

sequential problem

```
In [2]: a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = a
a = b
b = c
print("first no is now ",a,"second no. is",b)
```

first no is now 2 second no. is 1

```
In [3]: a = 10
b = 7
x = a+b
y = a-b
c = a*b
d = a/b
e = a//b
f = a%b
g = a**b
print(x,y,c,d,e,f,g)
```

17 3 70 1.4285714285714286 1 3 10000000

```
In [4]: a = 10
b = 7
c = a&b
d = a|b
e = ~a
f = a^b
g = a>>2
h = a<<2
print(c,d,e,f,g,h)
```

2 15 -11 13 2 40

```
In [5]: a = 10
b = 7
c = a>b
d = a<b
e = a==b
```

```
f = a!=b
g = a>=b
h = a<=b
print(c,d,e,f,g,h)
```

True False False True True False

```
In [7]: a = int(input("Enter first number: "))
        b = int(input("Enter second number: "))

        print("\nArithmetic Operations")
        print("Addition =", a + b)
        print("Subtraction =", a - b)
        print("Multiplication =", a * b)
        print("Division =", a / b)
        print("Floor Division =", a // b)
        print("Modulus =", a % b)
        print("Power =", a ** b)

        print("\nLogical Operations")
        x = True
        y = False
        print("x and y =", x and y)
        print("x or y =", x or y)
        print("not x =", not x)
        print("a > b and a < b =", (a > b and a < b))
        print("a == b or a != b =", (a == b or a != b))
        print("not(a > b) =", not(a > b))

        print("\nBitwise Operations")
        print("a & b =", a & b)
        print("a | b =", a | b)
        print("a ^ b =", a ^ b)
        print("~a =", ~a)
        print("a << 1 =", a << 1)
        print("a >> 1 =", a >> 1)
```

```
Arithmetic Operations
Addition = 9
Subtraction = -5
Multiplication = 14
Division = 0.2857142857142857
Floor Division = 0
Modulus = 2
Power = 128
```

```
Logical Operations
x and y = False
x or y = True
not x = False
a > b and a < b = False
a == b or a != b = True
not(a > b) = True
```

```
Bitwise Operations
a & b = 2
a | b = 7
a ^ b = 5
~a = -3
a << 1 = 4
a >> 1 = 1
```

```
In [6]: num = int(input("Enter a 3 digit no.: "))
rev = (num % 10) * 100 + ((num // 10) % 10) * 10 + (num // 100)
print("Reversed no.", rev)
```

Reversed no. 321

```
In [8]: a = input();
x,y,z= a[0],a[1],a[2]
rev = z+y+x
print(rev)
```

321

```
In [9]: a = 10
b = 7
c = a-(-b)
print("sum is",c)
```

sum is 17

```
In [11]: import math
print("1. area of Triangle")
print("2. area of Circle")
print("3. area of Rectangle")
choice = int(input("ENTER CHOICE"))
if choice == 1:
    r = int(input("enter radius"))
    circle = 3.14 * r * r
    print(circle)
elif choice == 2:
```

```

h = int(input("enter height"))
b = int(input("enter base"))
triangle = 1/2 * b * h
print(triangle)
elif choice == 3:
    a = int(input("enter height"))
    b = int(input("enter breadth"))
    rectangle = a*b
    print(rectangle)
else:
    print(0)

```

1. area of Triangle
 2. area of Circle
 3. area of Rectangle
- 3.0

branching

```

In [13]: a, b, c = map(int, input("Enter 3 numbers: ").split())
middle = a + b + c - max(a, b, c) - min(a, b, c)
print("Middle number =", middle)

```

Middle number = 3

```

In [12]: a = 10
b = 2
c = 4
if a>b & a>c:
    print("max is",a)
elif b>c & b>a:
    print("max is",b)
else:
    print("max is",c)

```

max is 4

```

In [13]: a = 10
b = 2
c = 4
d = a if a<b and a<c else c
print(c)

```

4

```

In [18]: a = float(input("Enter side 1: "))
b = float(input("Enter side 2: "))
c = float(input("Enter side 3: "))
if a + b > c and a + c > b and b + c > a:
    print("It is a valid triangle.")
else:
    print("Not a valid triangle.")

```

Not a valid triangle.

```
In [15]: a = float(input("enter side 1: "))
b = float(input("enter side 2: "))
c = float(input("enter side 3: "))
if a + b > c and a + c > b and b + c > a:
    if a == b == c:
        print("equilateral triangle")
    elif a == b or b == c or a == c:
        print("isosceles triangle")
    elif a*a + b*b == c*c or a*a + c*c == b*b or b*b + c*c == a*a:
        print("right-angled triangle")
    else:
        print("scalene triangle")
else:
    print("Not a valid")
```

scalene triangle

```
In [ ]: ## looping
```

```
In [20]: #12
num = int(input("Enter a number: "))
s = 0
while num > 0:
    s += num % 10
    num //= 10
print("Sum of digits =", s)
```

Sum of digits = 6

```
In [18]: #13
n = int(input("enter no.:"))
c = 0
for i in range(1, n + 1):
    if n % i == 0:
        c += 1

if c == 2:
    print("prime")
else:
    print("not prime")
```

not prime

```
In [19]: #14
n = int(input("enter no.:"))
temp = n
s = 0

while n > 0:
    d = n % 10
    f = 1
    for i in range(1, d + 1):
        f *= i
    s += f
```

```

n //= 10

if s == temp:
    print("strong number")
else:
    print("not strong")

```

not strong

```

In [20]: #15
n = int(input("enter no.:"))
a = 0
b = 1
f = False

while a <= n:
    if a == n:
        f = True
        break
    c = a + b
    a = b
    b = c

if f:
    print("fibonacci")
else:
    print("not fibonacci")

```

not fibonacci

```

In [25]: #16
start = int(input("enter start no.:"))
end = int(input("enter end no.:"))
for i in range(start, end + 1):
    if i % 2 != 0:
        print("odd no. within a raneg",i)

```

odd no. within a raneg 3
 odd no. within a raneg 5
 odd no. within a raneg 7
 odd no. within a raneg 9

```

In [30]: #17
n = int(input("enter no. "))
def isprime(x):
    if x < 2:
        return False
    c = 0
    for i in range(1, x + 1):
        if x % i == 0:
            c += 1
    return c == 2
rev = 0
temp = n
while temp > 0:

```

```

    d = temp % 10
    rev = rev * 10 + d
    temp //= 10

    if isprime(n) and isprime(rev):
        print("twisted prime")
    else:
        print("not twisted prime")

```

not twisted prime

```

In [29]: #18
n = int(input("enter no. to check"))
temp = n
rev = 0

while n > 0:
    d = n % 10
    rev = rev * 10 + d
    n //= 10

    if rev == temp:
        print("palindrome")
    else:
        print("not palindrome")

```

palindrome

```

In [28]: #19
n = int(input("enter no. "))
temp = n
s = 0

while n > 0:
    d = n % 10
    s += d**3
    n //= 10

    if s == temp:
        print("armstrong")
    else:
        print("not armstrong")

```

not armstrong

```

In [26]: #20
h = input("enter hex no.")
print("decimal =", int(h, 16))

```

decimal = 10

```

In [27]: #20
n = int(input("enter dec no. "))
print("hex =", hex(n))

```

hex = 0x511c0